

- 1 (a) The opening and closing of stomata involves the movement of potassium ions into and out of guard cells in response to environmental factors, for example light and water stress.

This movement of potassium ions is influenced by the concentration of calcium ions in the cytoplasm of guard cells. When plants are water stressed, abscisic acid (ABA) release causes the concentration of calcium ions in the cytoplasm to increase. The increase in calcium ions causes a series of changes resulting in the loss of potassium ions, so the stomata close in light.

A student investigated the effect of calcium chloride in overcoming the effect of light on stomata.

The student was provided with:

- 500 cm³ of 1.0 mmol dm⁻³ buffered calcium chloride (CaCl₂) solution
- freshly picked leaves from a plant that had been kept in light for 2 hours so that all the stomata were open.

The student soaked three epidermal strips of leaves in each of a range of calcium chloride solutions for 3 hours and recorded the number of closed stomata. An epidermal strip is made by peeling the epidermis from a leaf as a single layer.

- (i) Identify the independent and dependent variables in this investigation.

independent variable

.....

dependent variable

.....[2]

- (ii) The student used the 1.0 mmol dm⁻³ CaCl₂ solution provided to prepare 50 cm³ of four other concentrations by reducing the concentration by half each time.

State the method that the student used to make these four concentrations.

.....

.....

.....[1]

- (b) (i) Suggest a hypothesis that the student could test about the effect of CaCl₂ on the opening and closing of stomata.

.....

.....

.....[1]

(c) The student also tested the hypothesis:

The stomata on the upper epidermis of a leaf and the stomata on the lower epidermis respond differently to ABA.

- Young leaves that had just expanded were removed from seedling plants and cut into strips 5 mm wide and 10 mm long.
 - The upper and lower epidermis were removed from five leaf strips and put into the same concentration of buffered ABA solution.
 - This process was repeated for each different concentration of ABA solution.
 - These were left in a constant environment and supplied with carbon dioxide-free air for two hours.
 - The diameter of each stomatal aperture was measured using an eyepiece graticule.
 - The eyepiece graticule was calibrated using a stage micrometer at $\times 400$ and the stomata measured at the same magnification using the same microscope.
 - 10 randomly selected stomata on each of 5 epidermal strips from the lower and upper epidermis were measured at each ABA concentration.
- (i) State two ways in which the method of measuring the diameter of the stomata has been standardised.

1.
2.[1]

Fig. 1.1 shows stomata at different stages of opening.

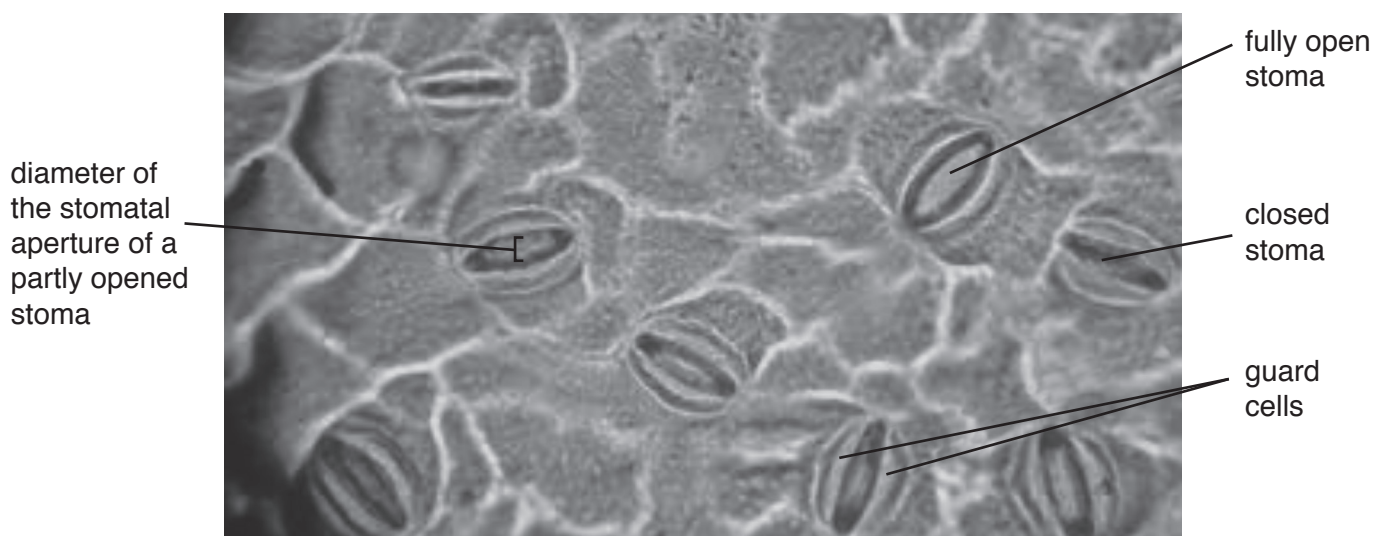


Fig. 1.1

(ii) Fig. 1.2 shows the eyepiece graticule and stage micrometer used by the student.

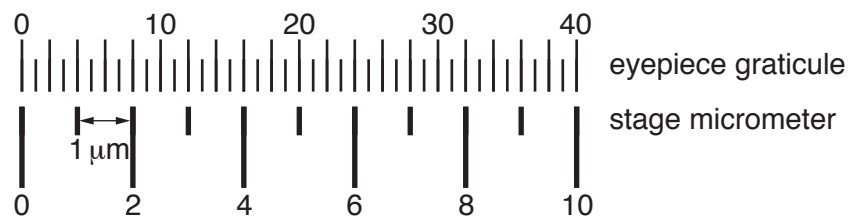


Fig. 1.2

The diameter of a stomatal aperture was 3 eyepiece units.

Calculate the actual size of the aperture.

..... μm [1]

(d) Fig. 1.3 shows the results of the experiment. The 95% confidence limits are shown for each mean stomatal aperture.

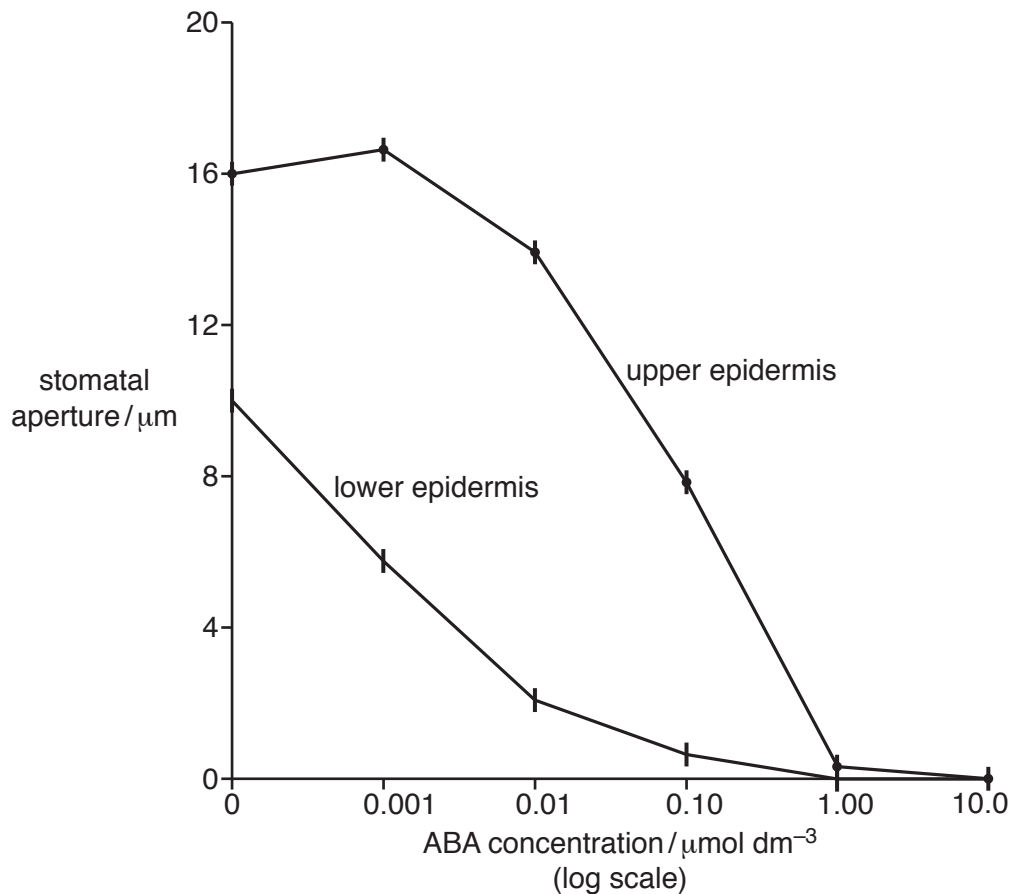


Fig. 1.3

(i) State **one** piece of evidence in Fig. 1.3 that supports the student's hypothesis that:

The stomata on the upper epidermis of a leaf and the stomata on the lower epidermis respond differently to ABA.

.....

-[1]

- Give a reason for your choice.

test

reason[2]

- The stomata on the upper epidermis of a leaf and the stomata on the lower epidermis respond differently to ABA.

[4]

[Total:19]